

JAYANT DETHE

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EDUCATION

Vidyalankar Institute of Technology

B.Tech in Information Technology with Specialization in AI/ML (Honors) – CGPA: 8.9

2022 – 2026

Mumbai, India

EXPERIENCE

Aivora Solutions

August 2025 – Feb 2026

Software Engineer — AI Infrastructure (Remote)

Mumbai, India

- Architected and deployed a production-grade FastAPI backend serving AI inference workloads; reduced average API response latency by 35 through async request handling and connection pooling.
- Designed a multi-tenant prompt orchestration layer supporting 3 concurrent product lines; decoupled business logic from LLM provider using an adapter pattern, cutting provider-switch cost from days to hours.

PROJECTS

StreamBill Pro 🚀 | [Python](#), [FastAPI](#), [PostgreSQL](#), [Redis Streams](#), [Celery](#), [Docker](#)

March 2026

- Engineered a multi-tenant usage-based billing and API metering platform; implemented graduated tier pricing engine supporting configurable pricing tiers with flat fees, volume discounts, and overage billing across multiple organizations.
- Built a high-throughput event ingestion pipeline using Redis Streams with consumer groups enabling guaranteed at-least-once delivery, automatic redelivery on worker failure, and batch DB writes of 100 events per flush — decoupling ingestion from storage under burst traffic.
- Implemented idempotent event deduplication via Redis, outbox pattern for crash-safe invoice delivery, WebSocket-based live usage dashboard, and Celery workers for hourly aggregation and payment retry with exponential backoff.

IntelliFix Engine 🚀 | [Python](#), [LangGraph](#), [Docker](#), [GitHub API](#), [Multi-Agent Systems](#)

January 2026

- Engineered a fault-tolerant autonomous code-repair system: a 3-agent LangGraph pipeline (analyser → fixer → validator) that reads GitHub issues, generates patches, and raises PRs with zero human intervention.
- Implemented Docker-sandboxed execution with iterative retry-backoff logic; isolated environment state per run to eliminate cross-execution contamination — reduced false-failure rate by 60 percent vs stateless execution.
- Built a dynamic entry-point detection algorithm using AST parsing across multi-file repositories; system successfully resolved issues on real-world OSS repos averaging 200+ source files.

URL Shortener 🚀 | [Node.js](#), [Express](#), [MongoDB Atlas](#), [Redis](#), [Docker](#), [Render](#)

December 2025

- Built a production-deployed URL shortening service using Base62 encoding over a 62^6 keyspace (~56B combinations); implemented optimistic locking in MongoDB to prevent key collisions under concurrent write load.
- Reduced redirect latency by ~82% using Redis write-through caching with TTL aligned to URL expiry; validated via load test at 300 concurrent requests — p95 latency dropped from 190ms to 34ms.
- Designed async analytics writes via an event-queue pattern to decouple redirect path from DB writes; enforced sliding-window rate limiting (100 req/min/IP) and containerised with Docker for deployment on Render.

TECHNICAL SKILLS

- Languages:** Python (advanced), Java, JavaScript / TypeScript, Go
- Backend & APIs:** Node.js, FastAPI, Express.js, REST API design, async programming, WebSockets
- Infrastructure & DevOps:** Docker, AWS (EC2, S3), CI/CD (GitHub Actions, Jenkins), Linux, Render, Vercel
- Databases & Caching:** PostgreSQL, MongoDB Atlas, Redis (caching, rate limiting), MySQL
- AI / ML Stack:** LLMs, RAG, LangChain, LangGraph, LlamaIndex, Hugging Face, PyTorch, Fine-tuning (LoRA, PEFT)
- Tools & Practices:** Git, GitHub API, unit testing (pytest), system design, agile workflows

COMPETITIVE PROGRAMMING & DSA

- LeetCode:** 250+ problems solved — core focus on graphs (BFS/DFS, Dijkstra), dynamic programming, sliding window, and binary search
- Problem Solving:** Consistent practice on time/space complexity analysis; comfortable with $O(n \log n)$ optimisation trade-offs in real system design contexts

CERTIFICATIONS

- Generative AI with LangChain & HuggingFace – Udemy
- Business Management Course – IIM Mumbai

- Junior Programmer Pathway – Unity
- TCS iON Career Edge - Young Professional